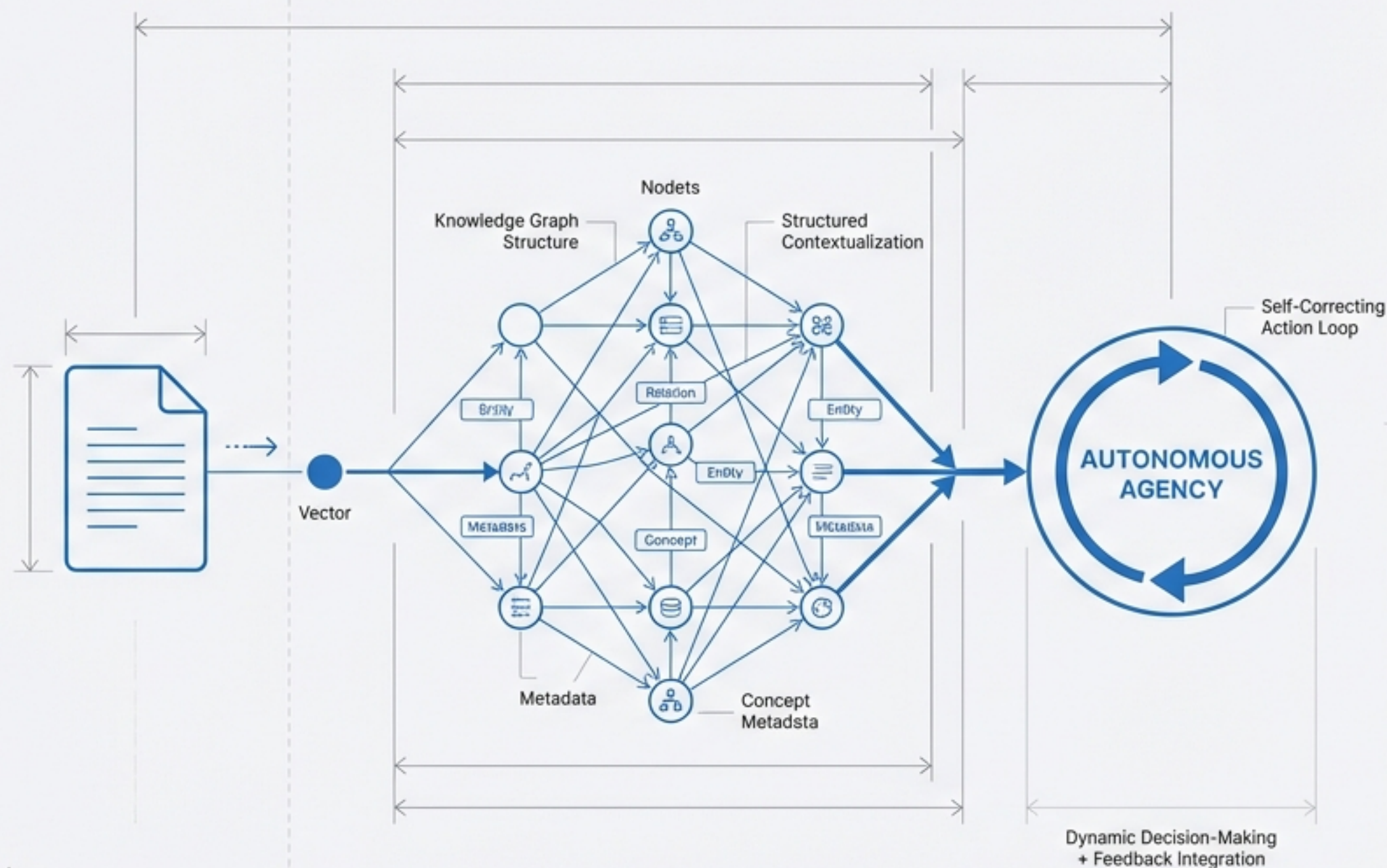


The RAG Maturity Curve

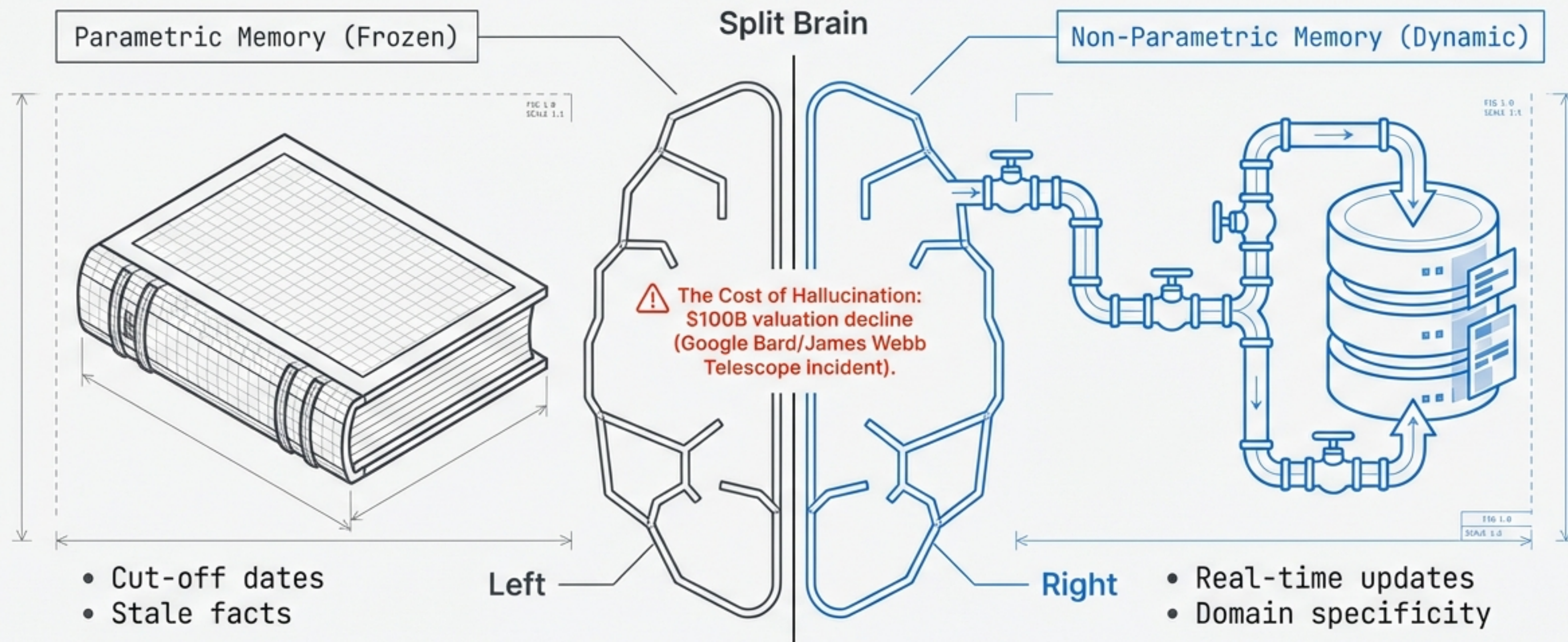
From Static Retrieval
to Autonomous Agency



JetBrains Mono

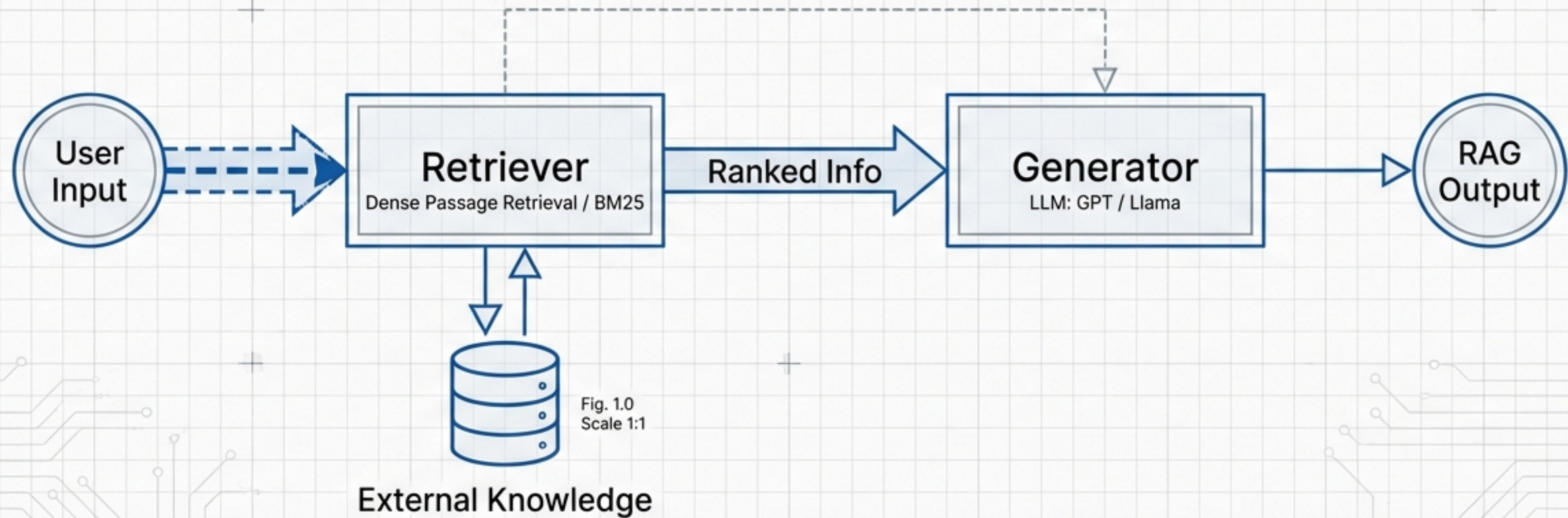
Architecture | Optimization | Security Protocols

The Gap Between Frozen Memory and Dynamic Reality



RAG blends generation with document lookup to stick to the facts, mitigating the risk of plausible but incorrect generation.

Anatomy of a Standard RAG Pipeline



Bi-Encoder Architecture: Queries and documents are encoded separately into high-dimensional vectors for efficient nearest-neighbor search.

The Art of Chunking: Optimization at the Ingestion Layer

Fixed-Size	Recursive	Semantic	Late Chunking
Simple token count. Risk: Cuts sentences in half, losing semantic meaning.	Respects paragraphs/newlines. Benefit: Maintains better context flow.	Splits by topic changes. Benefit: High coherence.	Embeds entire context first. Benefit: Preserves global context.

Visual Example: Before/After

Bad Chunking

Fig. 3.0
Scale 1:1

The RAG pipeline is a framework for
[CHUNK_BREAK] optimizing LLM output.

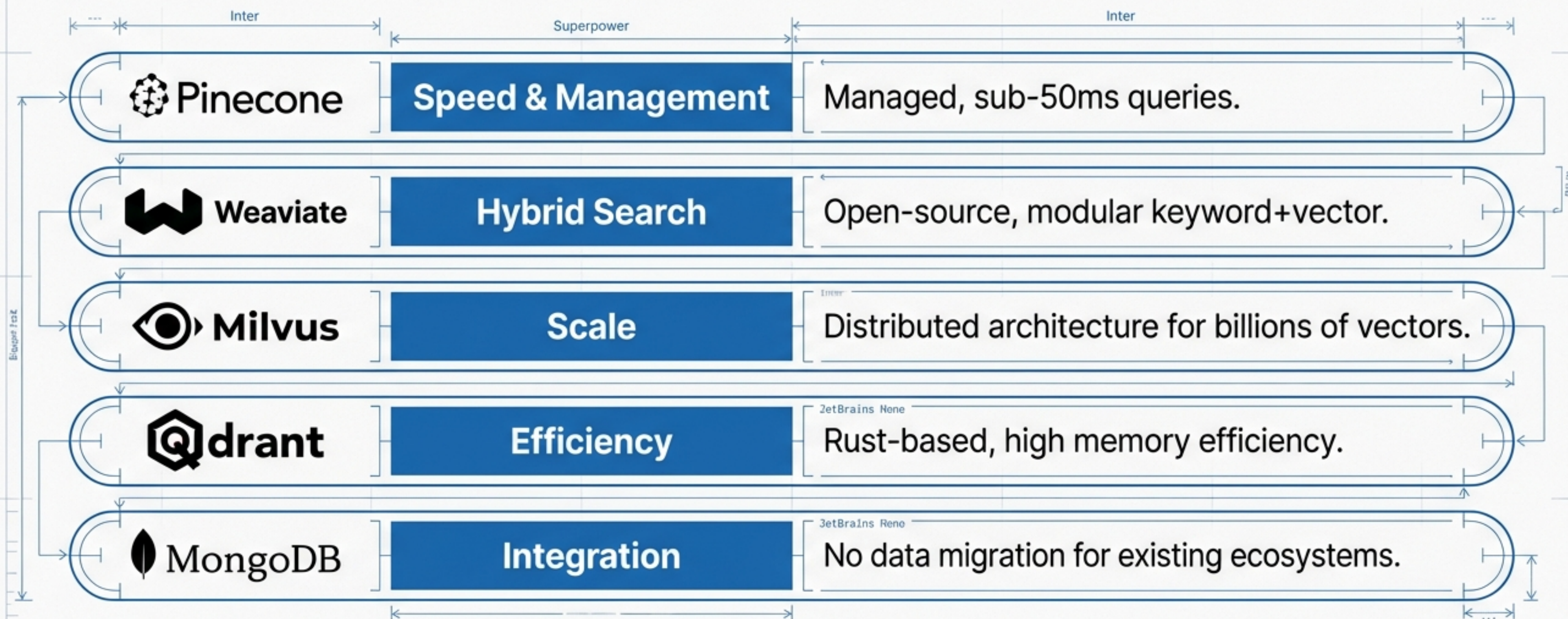
Good Chunking

Fig. 3.8
Scale 1:1

The RAG pipeline is a framework for
optimizing LLM output.

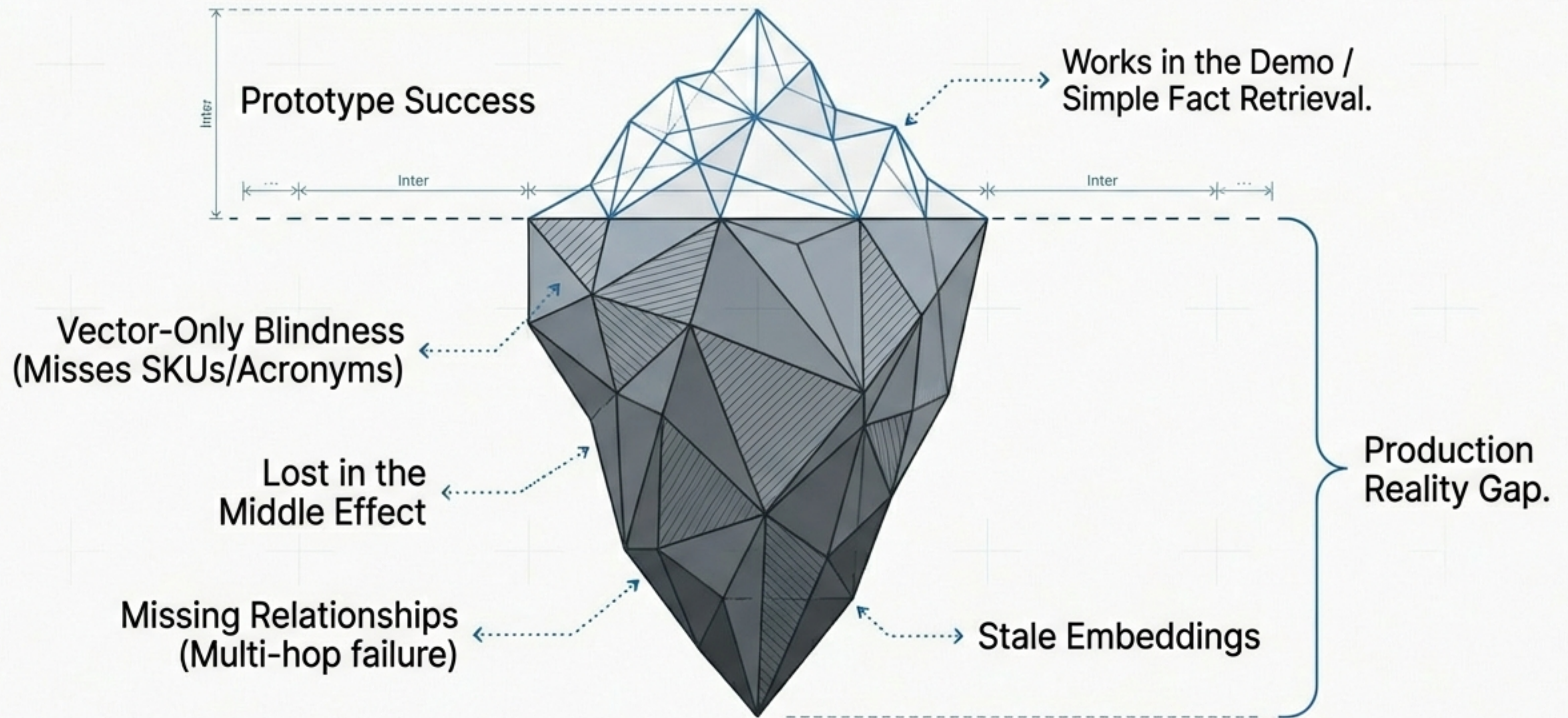
The Database Landscape

Fig. 3.0
Scale 1:1



Heuristic: Select based on latency vs. ecosystem integration vs. operational overhead.

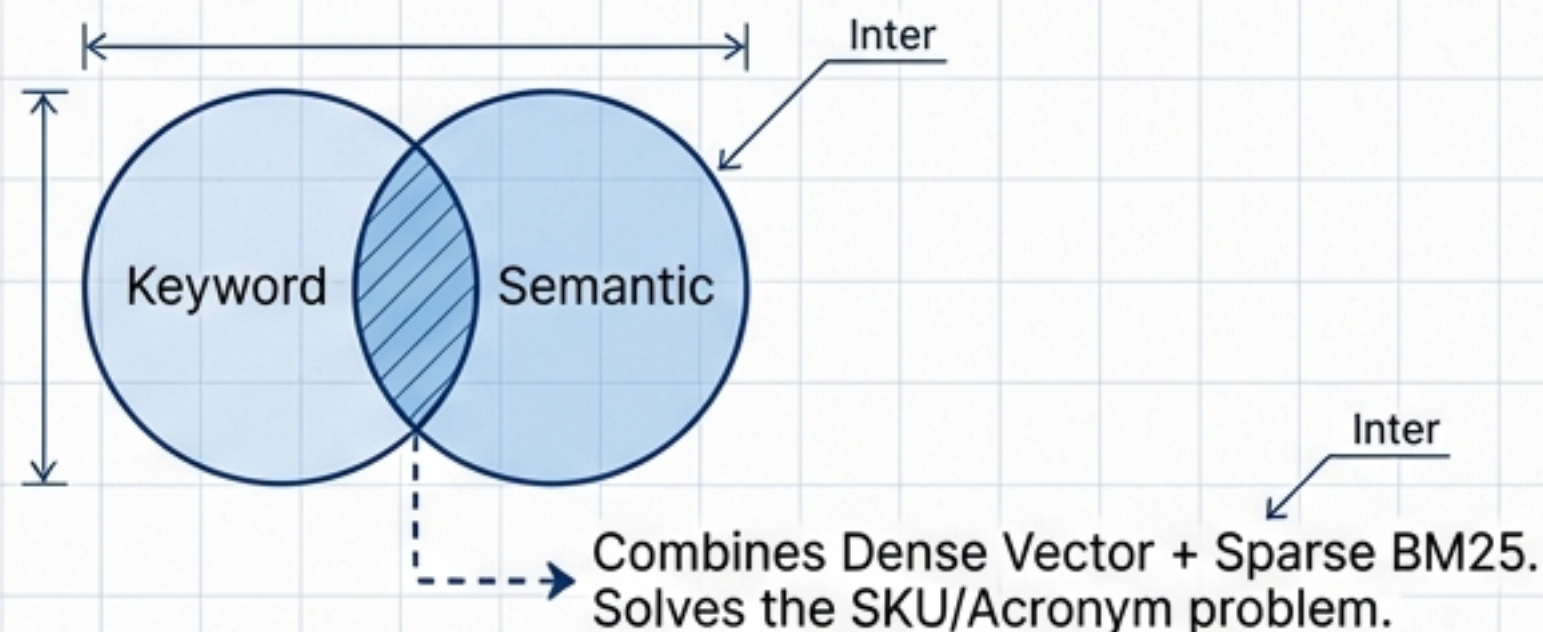
The Quality Ceiling: Why Basic RAG Breaks



Advanced Techniques to Breach the Quality Ceiling

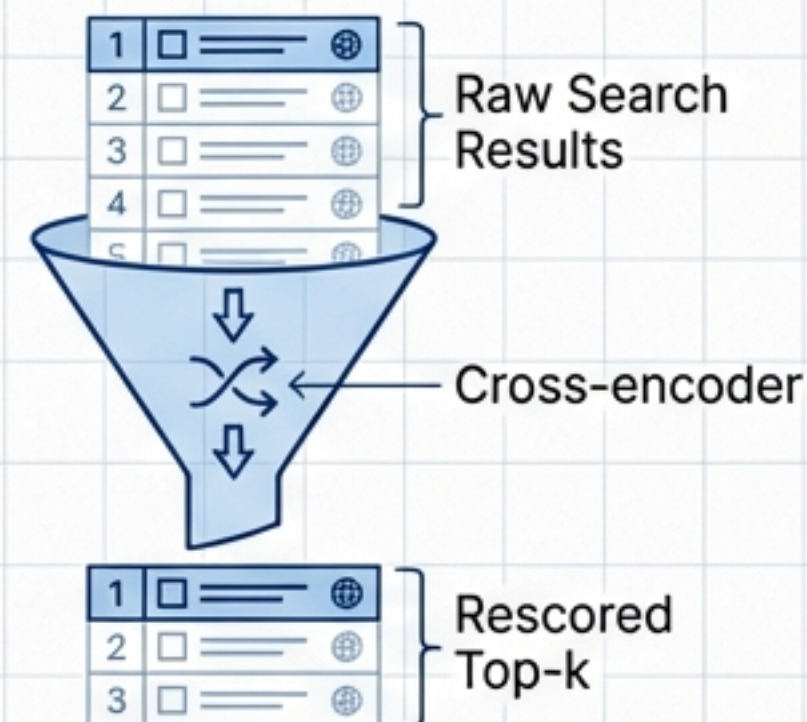
Fig 4.0
Scale 1:1

Hybrid Search

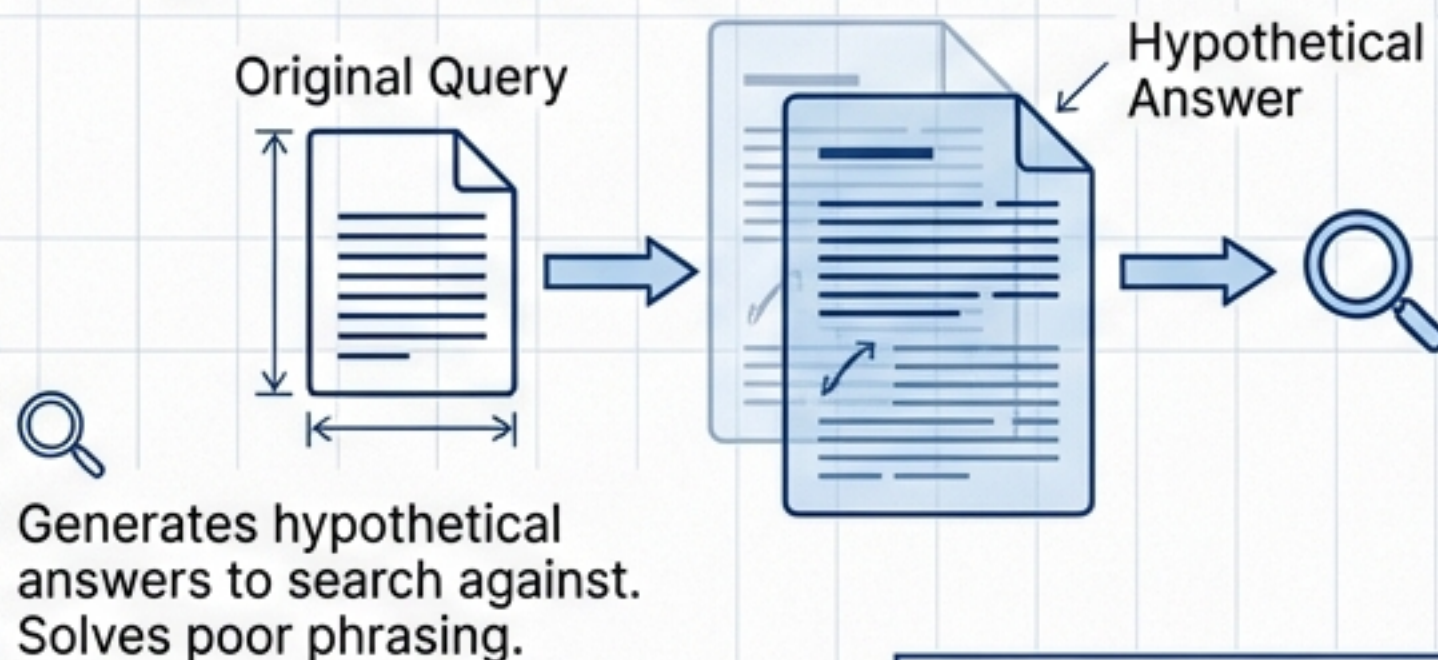


Reranking

 Cross-encoder re-scoring of top-k results. Maximizes precision.

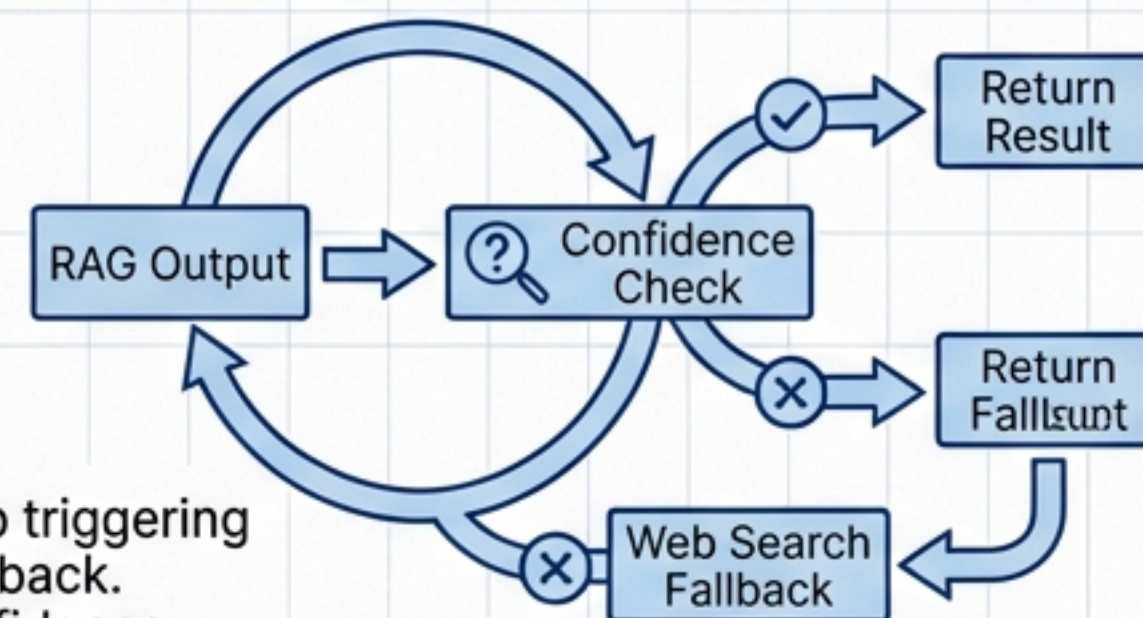


Query Expansion (HyDE)



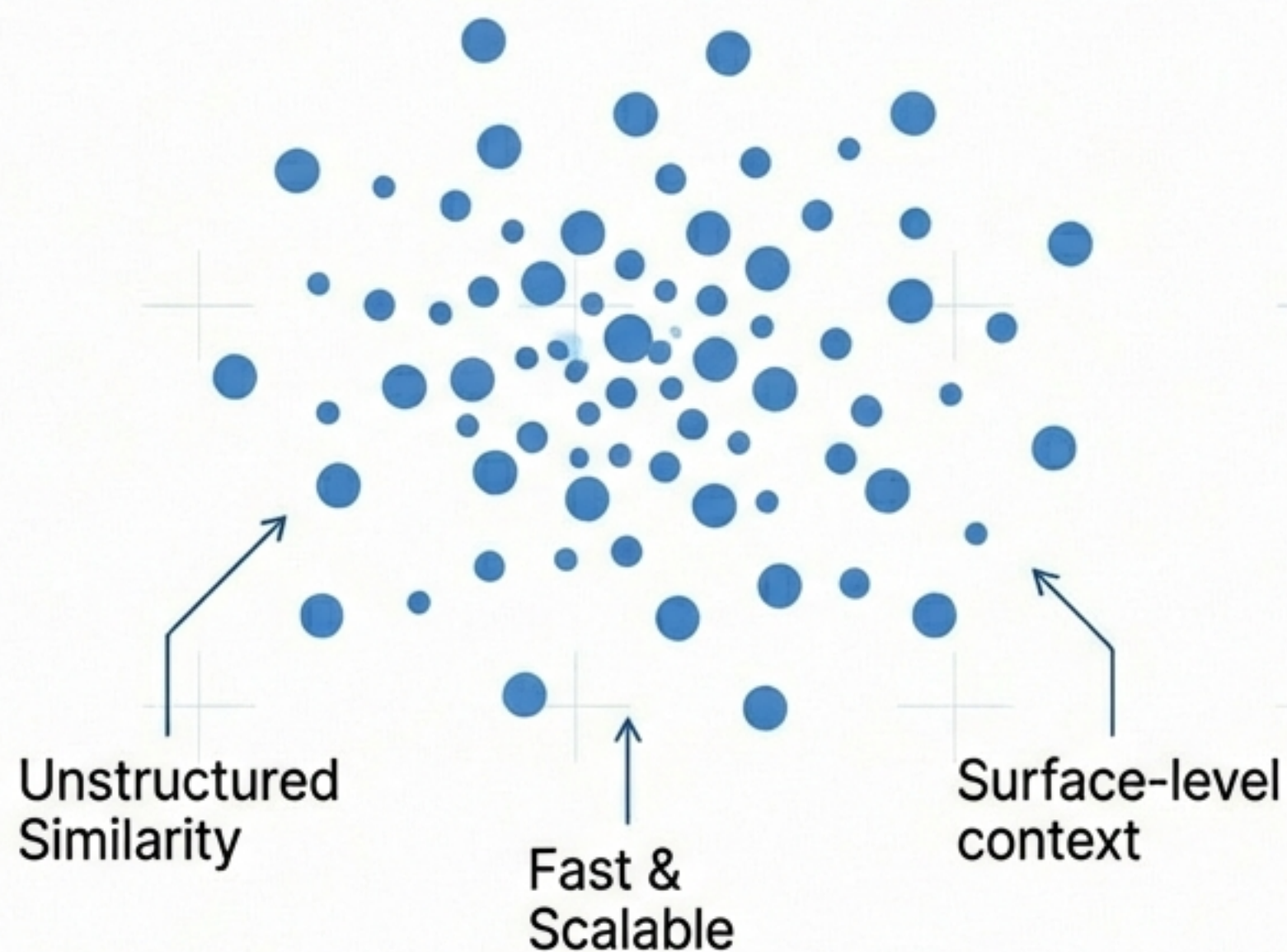
Corrective RAG (CRAG)

 Self-check loop triggering web search fallback. Solves low confidence.

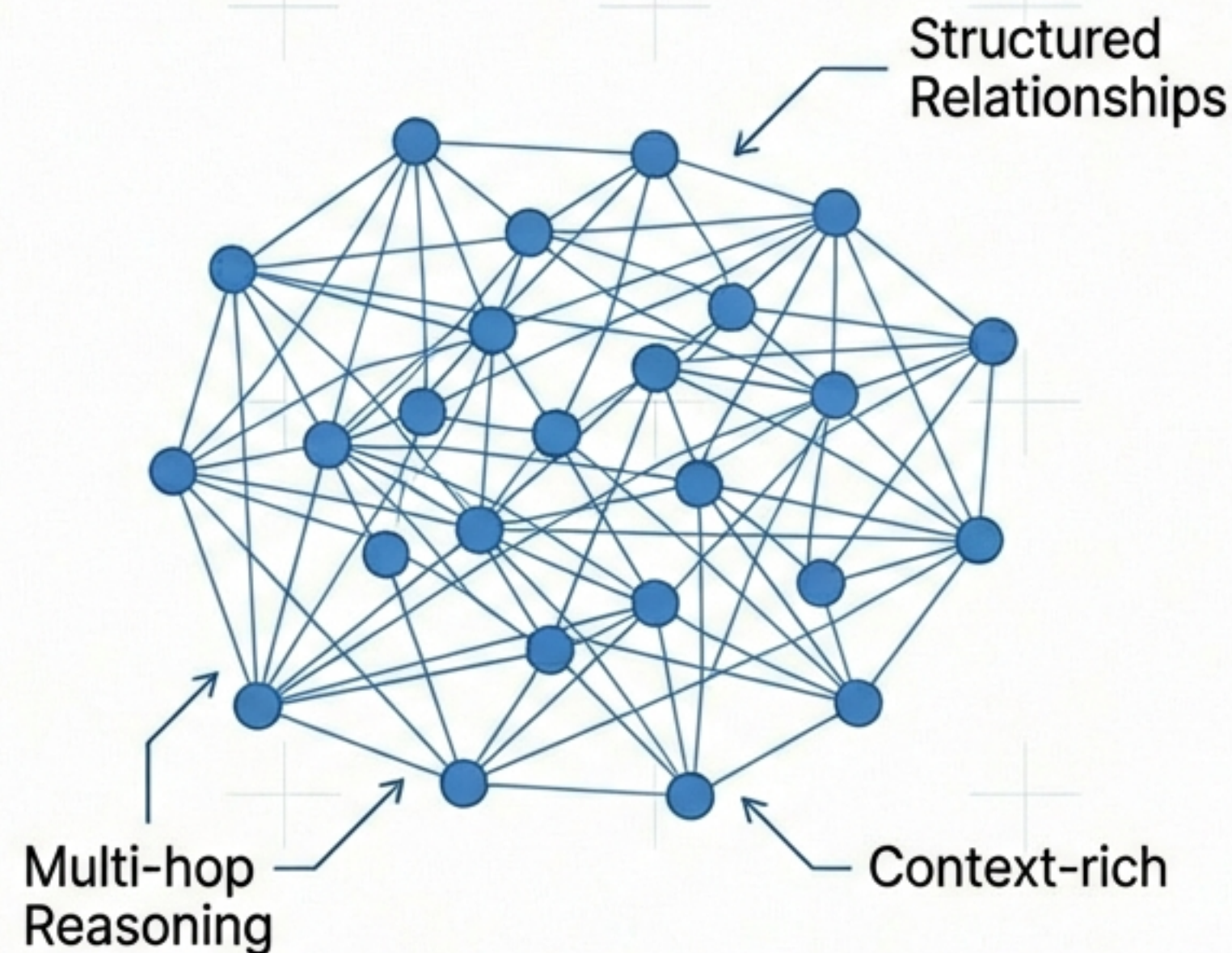


The Structural Shift: Vector RAG vs. GraphRAG

Vector RAG



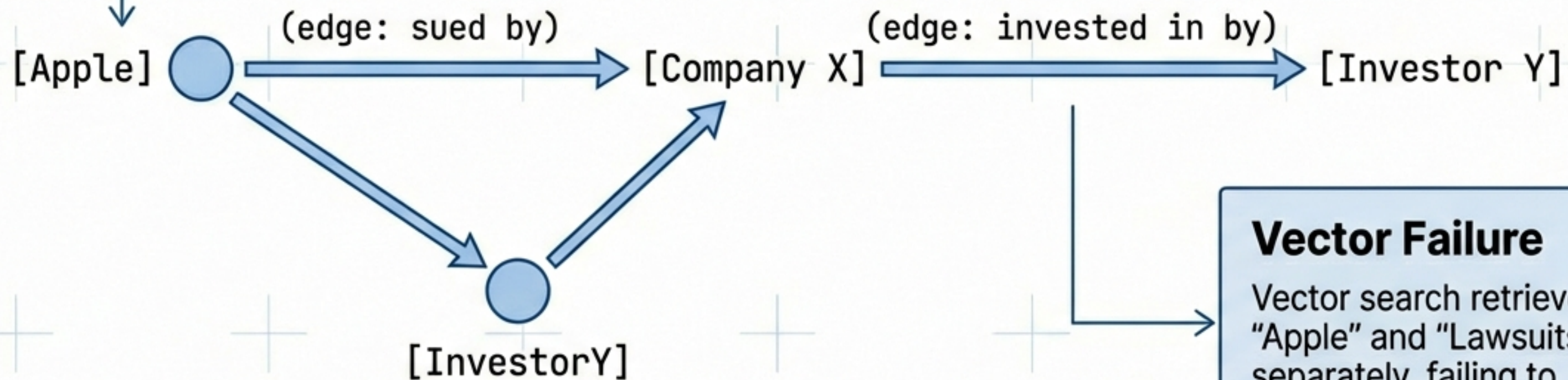
GraphRAG



When to Deploy Knowledge Graphs

Solving the Multi-Hop Reasoning Problem

Query: "Find the investors of companies that sued Apple in 2024."

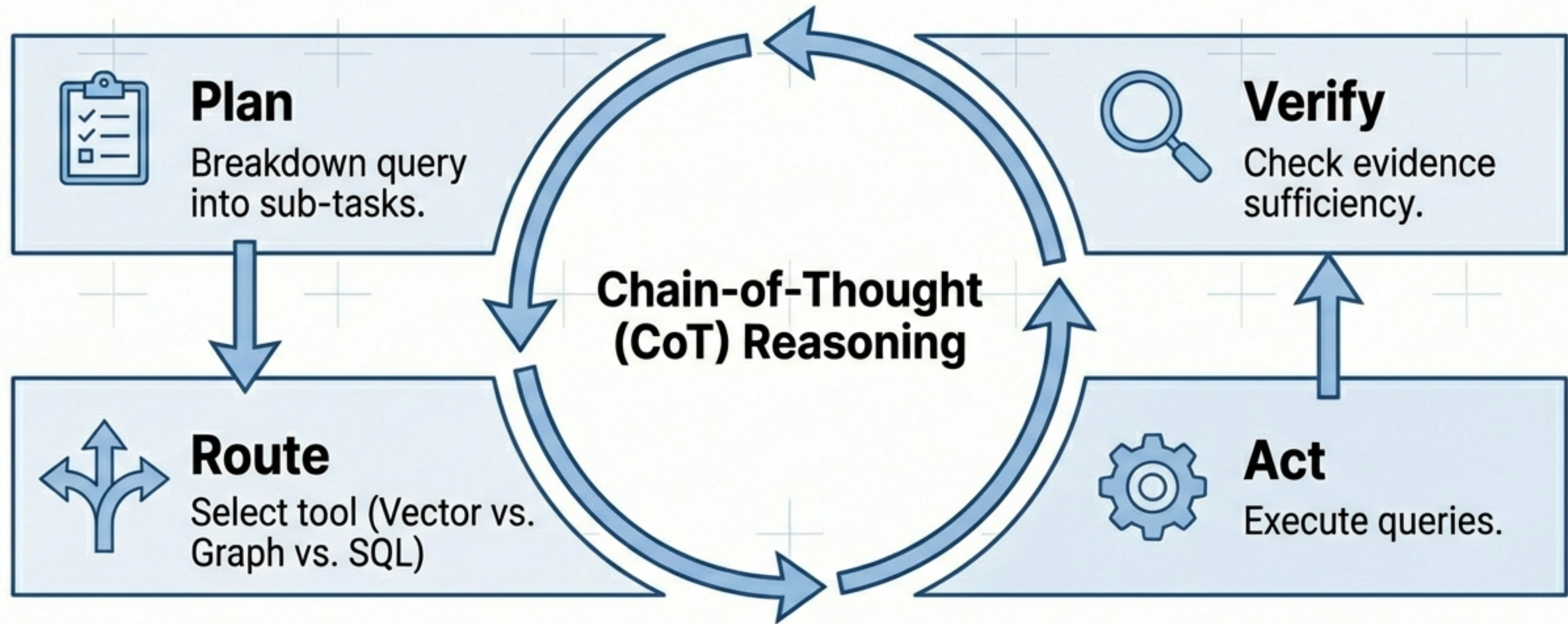


Vector Failure

Vector search retrieves "Apple" and "Lawsuits" separately, failing to bridge the gap.

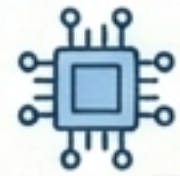
GraphRAG supports domain modeling, allowing traversal across entities to answer global, cross-document questions.

From Retrieval to Agency: The Agentic Loop



Agents move beyond simple fetching to dynamic reasoning and planning.

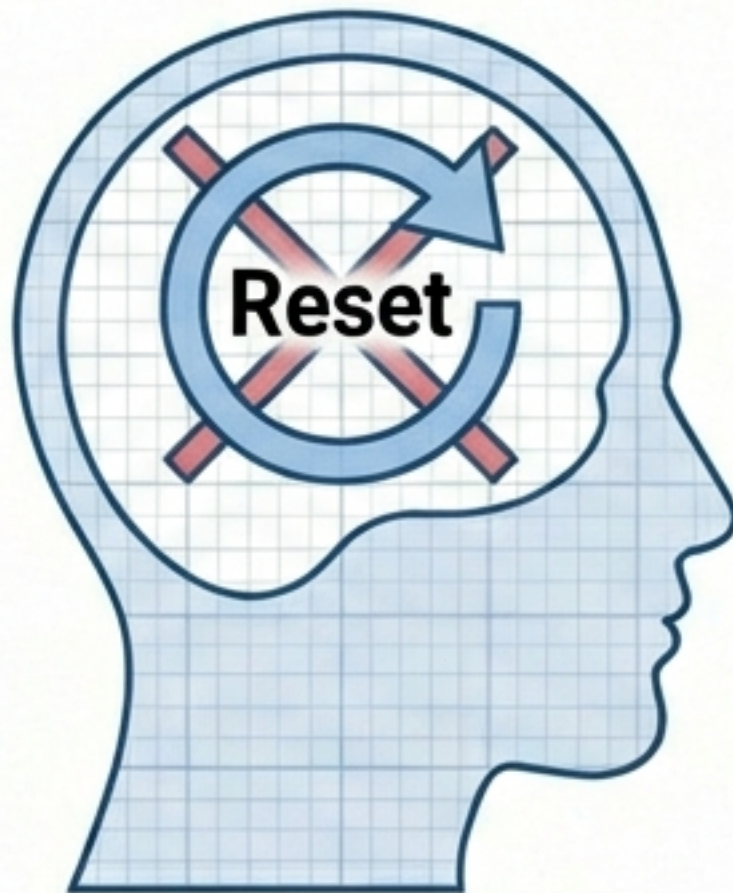
The Memory Layer: Solving the Stateless Problem



Solving the Stateless Problem

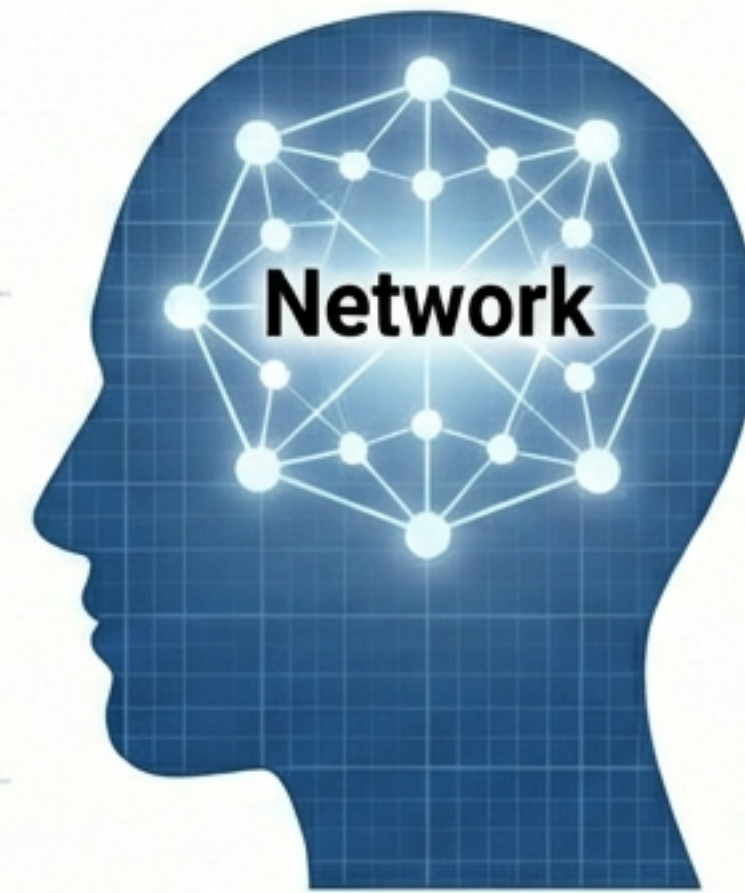
Traditional RAG (Amnesic)

Agentic Memory



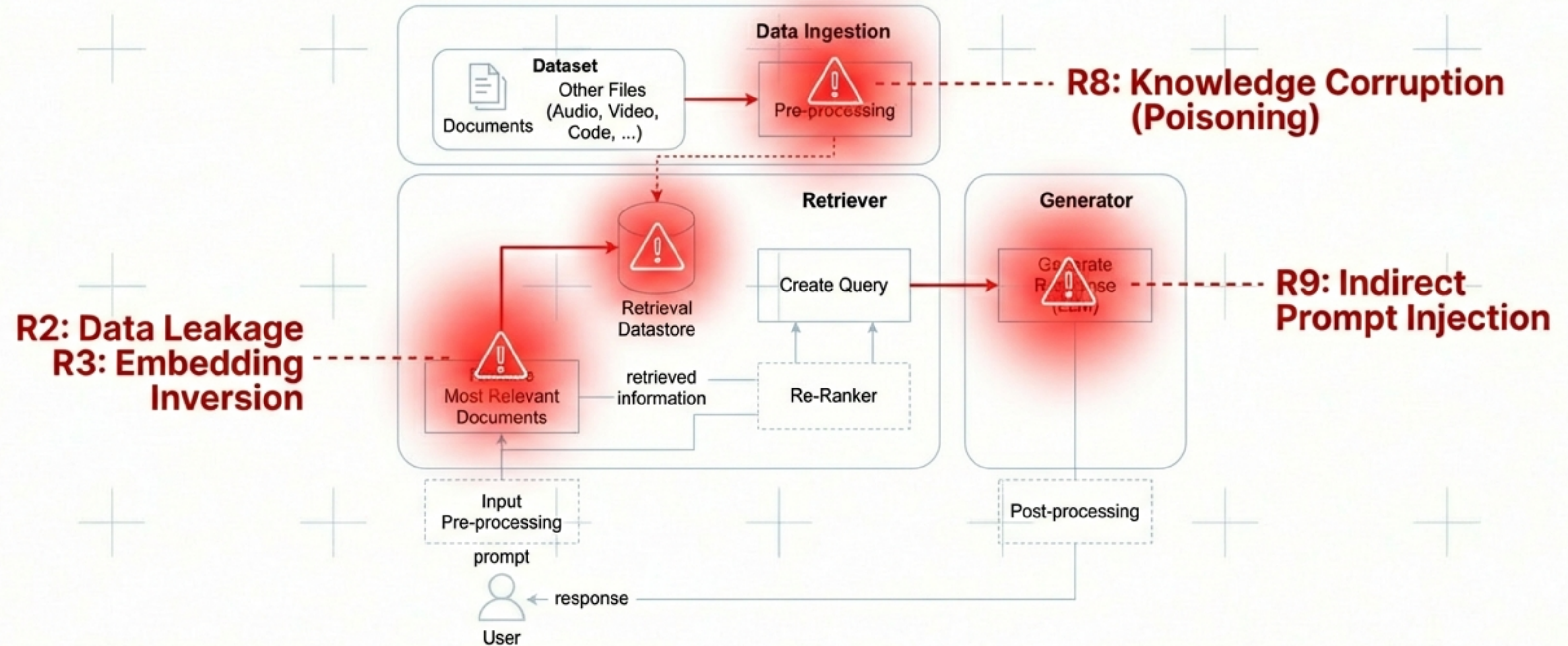
Session resets. High token cost from re-uploading context.

Memory compression cuts token usage by up to 80%.



Short-term (Working), Episodic (History), Semantic (Preferences).

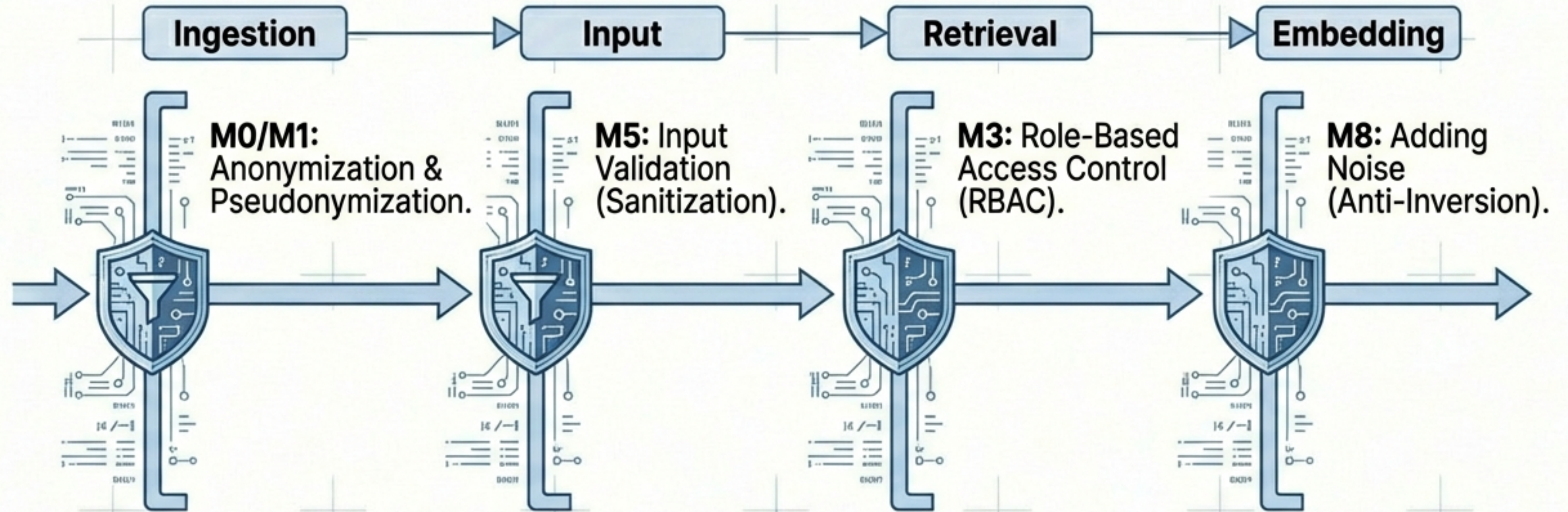
The RAG Attack Surface



New capabilities bring new vulnerabilities.

Source: arXiv Security Framework

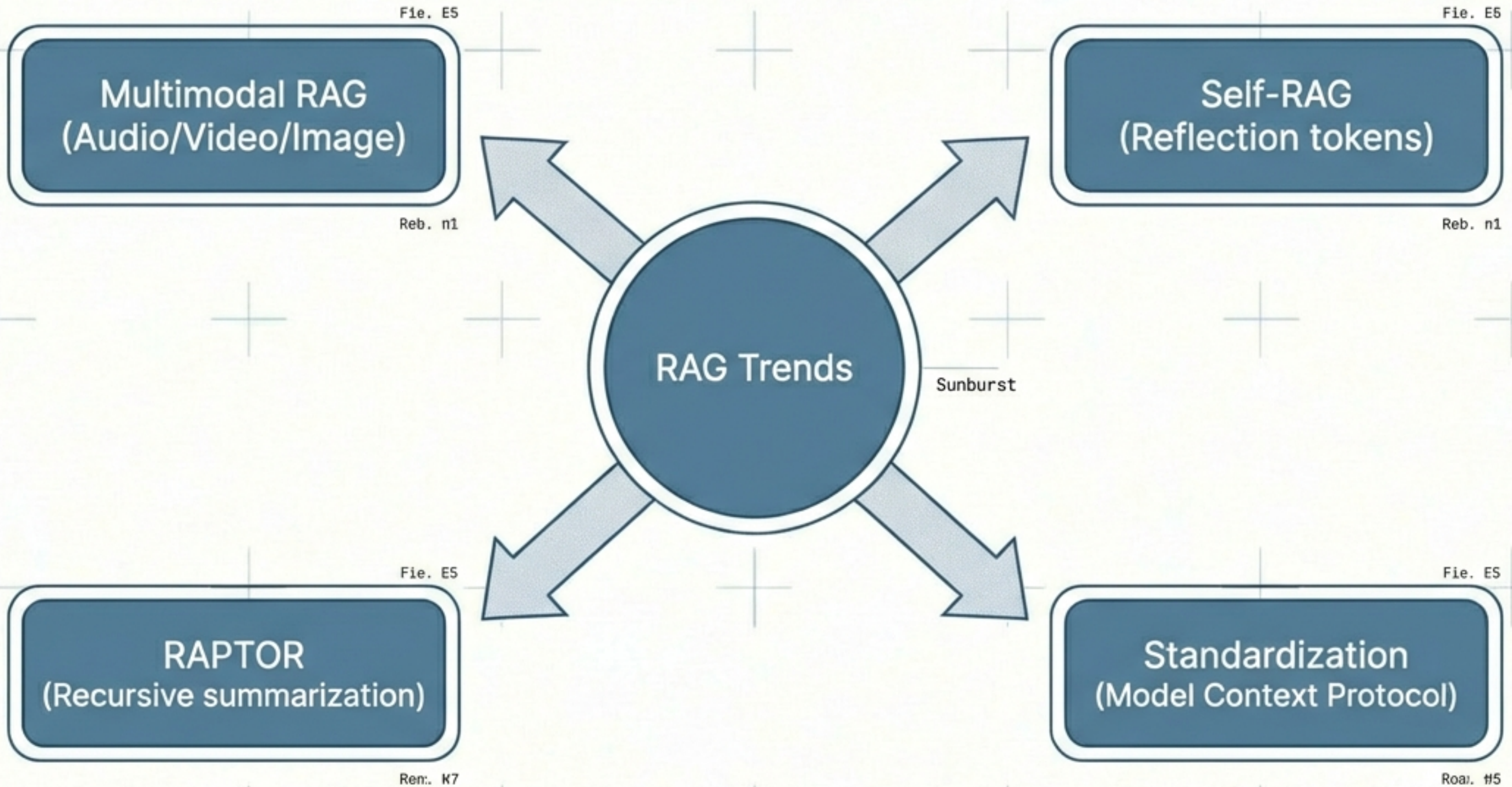
Security Mitigation Framework



Defense-in-depth strategy mapping to identified risks.

Source: arXiv Security Framework

Future Directions



Strategic Takeaways: The Maturity Checklist



Ingestion: Shift from fixed-size to Semantic or Late Chunking.



Structure: Evaluate GraphRAG for complex reasoning.



Autonomy: Implement Agentic loops with planning steps.



Memory: Deploy persistent memory to reduce token costs.



Security: Harden pipeline against injection and inversion.

“The future of retrieval is not just about finding data—it is about reasoning with it securely.”